PharmAchieve

COMMON ANEMIAS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

Providing Care: Clinical Care

Providing Care: Distribution

Knowledge & Expertise

Communication & Collaboration

Leadership & Stewardship

Professionalism

LEGEND

AA	Amino Acid
CO_2	Carbon Dioxide
CO	Carbon Monoxide
CRP	C-Reactive Protein
DNA	Deoxyribonucelic Acid
EPO	Erythropoietin
G6PD	Glucose-6-Phosphate Dehydrogenase
GERD	Gastroesophageal Reflux Disease
GI	Gastrointestinal
Hct	Hematocrit
Hgb	Hemoglobin
HIV	Human Immunodeficiency Virus
IDA	Iron Deficiency Anemia
MCH	Mean Corpuscular Hemoglobin
MCHC	Mean Cell Corpuscular Concentration
MCV	Mean Corpuscular Volume
O_2	Oxygen
RBCs	Red Blood Cells
RDW	Red Cell Distribution Width
TIBC	Total Iron Binding Capacity
TSAT	Transferrin Saturation
WHO	World Health Organization

COMPONENTS OF BLOOD

FYI

Fluid portion containing proteins such as albumin and globulin

Leukocytes (white blood cells)
- Attack diseased cells or invaders; increase in number indicates presence of disease
- <1% of total blood volume

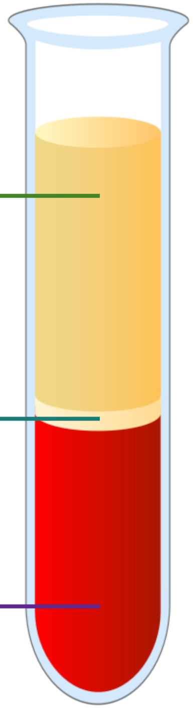
Thrombocytes (platelets)
- Blood clotting; life span ~8 days

Erythrocytes (red blood cells)
- Gas transport by hemoglobin; lack nucleus and organelles
- Normal hematocrit: males: 40.7-50.3%; female: 36.1-44.3%
- Life span ~120 days

Plasma
Water, proteins, nutrients, hormones, etc.

Buffy coat
White blood cells, platelets

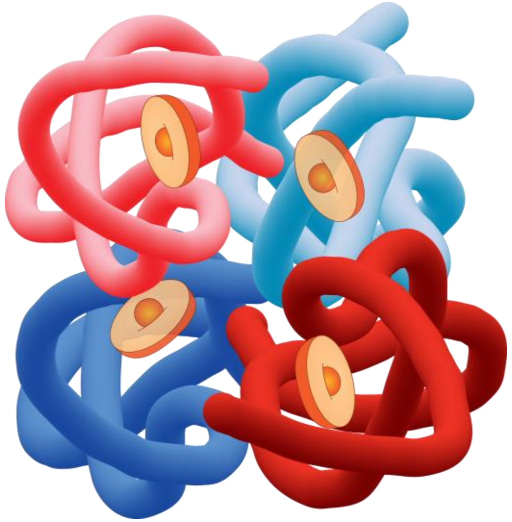
Hematocrit
Red blood cells

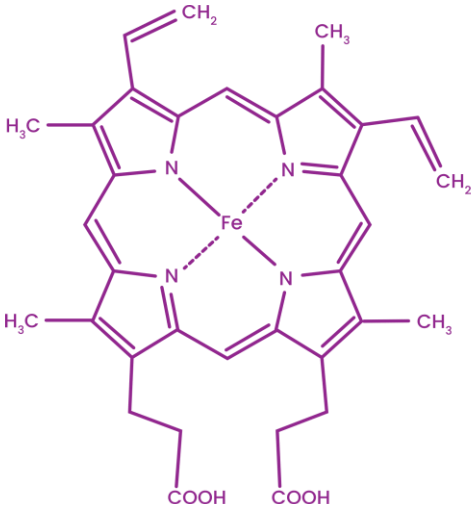


HEMOGLOBIN

FYI

- 4 globin subunits, each with a heme group: C-H-N porphyrin ring (made from glycine) with an Fe^{2+} in center
- O_2 distribution and CO_2 elimination: heme loosely binds to O_2 and CO_2
- Factors that decrease oxygen affinity: increased temperature, increased blood CO_2, decreased pH (increased blood H^+)
    - Tissues have higher values of these factors vs. the lungs in order to unload O_2
- Myoglobin transports O_2 in muscle tissues: a porphyrin monomer
- Carbon monoxide (CO) has a higher affinity for hemoglobin than oxygen – hemoglobin will be taken by CO during CO poisoning, depleting oxygen levels in the blood





ANEMIA

Collective term describing different blood conditions that decrease the oxygen carrying capacity

Results from decreased level of hemoglobin or RBCs

World Health Organization definition:
hemoglobin <120 g/L in non-pregnant females, hemoglobin <130 g/L in males

Clinical presentation common to all forms of anemia: dizziness, edema, fatigue, pale mucous membranes, pallor, severe congestive heart failure, shortness of breath, tachycardia, weakness

GOALS OF THERAPY

- ✓ Improve signs and symptoms of anemia
- ✓ Investigate and treat underlying cause(s)
- ✓ Restore normal hemoglobin levels
- ✓ Minimize side effects associated with drug therapy

PATHOPHYSIOLOGY

Hematopoiesis

- Maturation of blood components (granulocytes, lymphocytes, RBCs) from stem cells

Erythropoiesis

- Development of RBCs
- Erythropoietin (EPO) is the regulatory hormone required for RBC production
- Kidneys produce EPO in response to reduced oxygen levels
- EPO stimulates the bone marrow
- Normal development of RBCs requires adequate folate, vitamin B12, and iron
- It takes about one week for stem cells to mature into erythrocytes
- Mature RBCs have a lifespan of about 120 days

ERYTHROPOIESIS

FYI

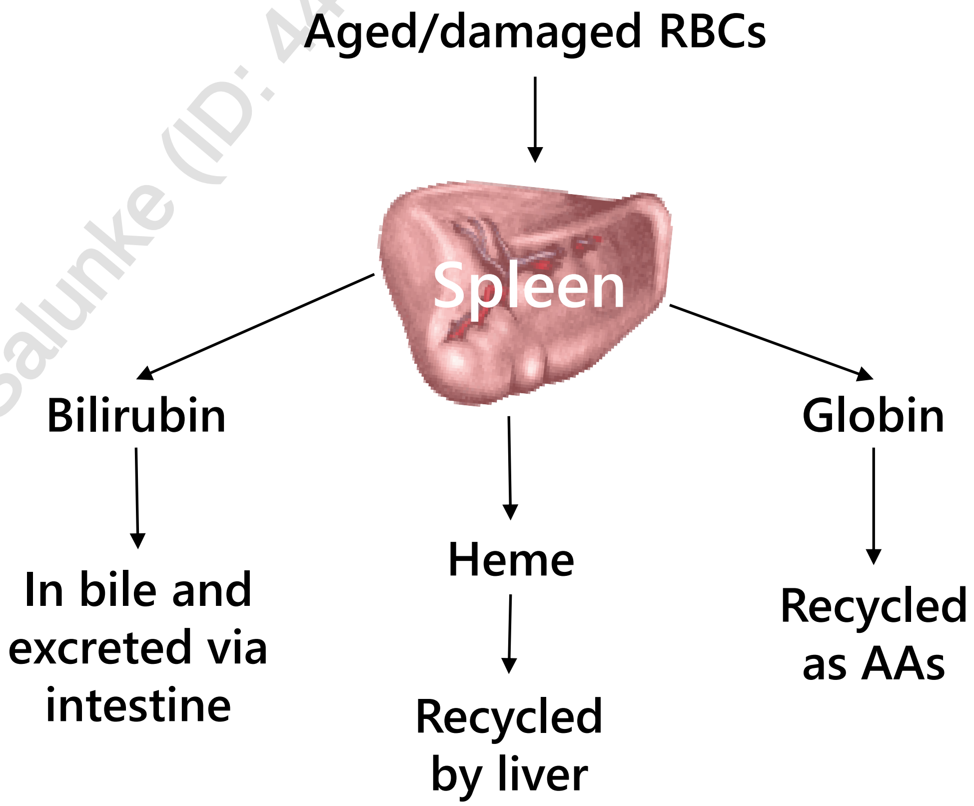
Hemocytoblast: uncommitted stem cell to all blood cells in the red marrow

Proerythroblast: committed to erythropoiesis, first detectable pre-cursor in RBC development

Erythroblast: has lots of ribosomes and hemoglobin

Reticulocyte: most organelles removed to reach concave shape and enters blood, matures in 1 day

Erythrocyte: ribosomes are degraded by intracellular enzymes



Erythropoietin (EPO) is a hormone produced in the kidneys that stimulates erythropoiesis; low O_2 levels trigger the kidneys to release erythropoietin

INTERPRETING LAB TESTS – DEFINITIONS

	Hemoglobin (Hgb)	Protein in Red Blood Cells (RBCs) that serves as a transporter of O2 to the organs and returns CO2 (de-oxygenated) blood back to the heart
	Hematocrit (Hct)	% (ratio) to determine the total volume of RBCs relative to the total volume of blood
	Reticulocyte count	% of immature RBCs (reticulocytes) produced and circulating compared to circulating (mature) erythrocytes; decreased counts indicate bone marrow suppression and increased counts indicate hemolytic anemia or acute blood loss
RBC Morphology	Mean Corpuscular Volume (MCV)	Hct/RBC count; average volume ("size") of RBCs: microcytic (small), normocytic (normal), or macrocytic (large)
RBC Morphology	Mean Corpuscular Hemoglobin (MCH)	Hgb/RBC count; average weight("amount") of Hgb in RBCs (i.e. hypochromic (decreased Hgb), normochromic (normal Hgb), or hyperchromic (increased Hgb)
RBC Morphology	Mean Cell Corpuscular Concentration (MCHC)	Hgb/Hct; average concentration of Hgb in RBCs

INTERPRETING LAB TESTS – DEFINITIONS

Serum iron

- Amount of iron bound to transferrin
- Transferrin delivers iron from absorption centers – iron is absorbed from the duodenum and is delivered to bone marrow (RBC precursors) and other organs through circulating blood; also incorporates iron into heme
  - Transferrin Saturation (TSAT) = %iron-binding sites used on transferrin = serum iron/ TIBC x 100%

Total Iron Binding Capacity (TIBC)

- Amount of iron that could bind to transferrin
- High TIBC + low serum iron = iron deficiency anemia

Ferritin

- Protein that serves to store iron in the tissues; levels are proportional to the total body iron stores; used to determine whether there is an excess or shortage of iron; small amount is found in circulating blood

INTERPRETING LAB TESTS – DEFINITIONS

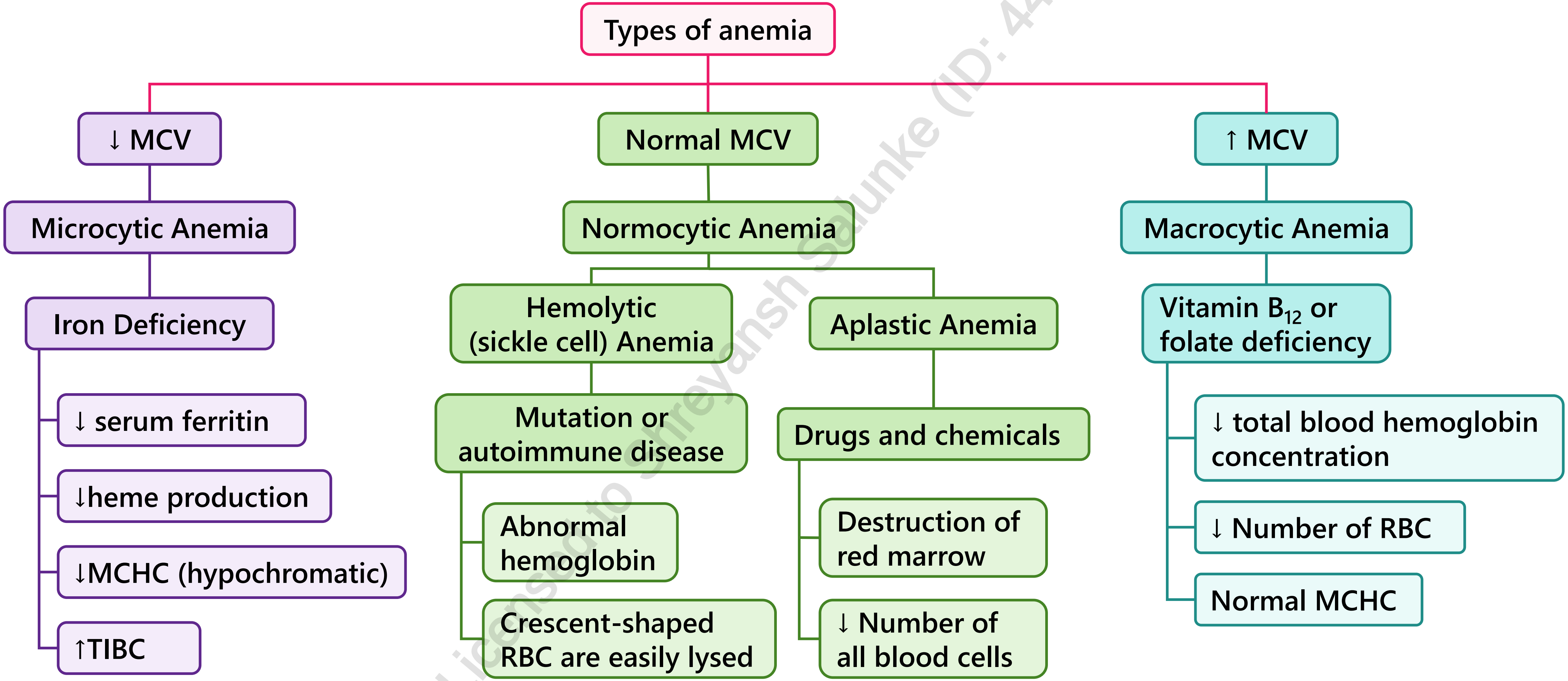
Vitamin B12 and folate

- Required in erythropoiesis to generate erythrocytes; involved in DNA synthesis and RBC maturation
- Increased folic acid levels may mask vitamin B12 deficiencies

Red Cell Distribution Width (RDW)

- The variation in RBC size
- If this number is large, it means cells are different sizes. For example, it would be high in macrocytic anemia, hemolytic anemia and iron deficiency anemia. It would be closer to normal in thalassemia
- Not sensitive or specific to make a diagnosis, this value is used in combination with others
- May be increased in anemias, venous thromboembolism, chronic kidney disease, liver disease

ANEMIA CLASSIFICATIONS



NORMOCYTIC ANEMIA

NORMOCYTIC ANEMIA = "NORMAL *size* but REDUCED # RBCs

Causes	• Hemolytic anemias • Anemia of chronic disease • Acute blood loss
Lab values	• Low hemoglobin or hematocrit • Normal MCV • Low or high reticulocyte counts

LOW reticulocyte counts

- Anemia of chronic disease

HIGH reticulocyte counts

- Acute blood loss
- Hemolytic anemias:
  - Spherocytosis
  - Sickle cell anemia
  - Other: G6PD deficiency, paroxysmal nocturnal hemoglobinuria, autoimmune hemolytic anemia

NORMOCYTIC ANEMIAS: HEMOLYTIC ANEMIAS

Spherocytosis

- Characterized by spherical RBCs (known as spherocytes)
- Hereditary condition
- Spherocyte membranes are unstable and inflexible which causes them to become trapped and destroyed by the spleen
- Treatment options: splenectomy (recommended), blood transfusions, EPO, folic acid (severe cases)

Sickle cell anemia

- Hereditary condition caused by amino acid substitution on the hemoglobin beta-chain resulting in sickle-shaped RBCs
- Symptoms: vaso-occlusion, leg ulcers, gall stones, splenic sequestration
- Treatment options: hydroxyurea, blood transfusions, pain management, avoidance of precipitants

NORMOCYTIC ANEMIA: ANEMIA OF CHRONIC DISEASE

- Also known as anemia of inflammation
- May co-exist with other anemias
- Commonly seen in hospitalized patients
- Associated with several disease states: HIV, autoimmune conditions, cancer, heart failure, chronic inflammation, renal disease
- No definitive test available, however common presentation includes:
    - High C-reactive protein
    - Normal or high ferritin levels
    - Low serum iron levels
    - Low TIBC
    - EPO deficiency
- Treatment: treat underlying cause
    - If iron deficient: supplement with iron
    - If not deficient in iron, folate, or B12: initiate erythropoiesis-stimulating agent

MICROCYTIC ANEMIA

Causes	• Iron Deficiency Anemia (IDA) • Thalassemia • Lead poisoning • Chronic disease

Occurs because of impaired hemoglobin synthesis

Lab values	• Low hemoglobin or hematocrit • Low MCV

Iron-deficient anemia

- Most common anemia and nutritional deficiency
- Lab values specific to IDA: low MCV, low serum iron, low ferritin levels, high TIBC
- Common causes: chronic blood loss (heavy menstruation, GI malignancy), chronic renal failure, inadequate intake, infancy and young children, poor absorption (malabsorption, H. pylori, celiac disease, gastric bypass, foods containing calcium, medications), pregnancy
- Symptoms specific to IDA: koilonychia (concave nails), pica (cravings for non-nutritive substances such as cornstarch, ice, and clay), atrophic glossitis (inflammation of the tongue), angular stomatitis (inflammation around the mouth)

Iron is a component of heme

Iron deficiency
↓
defective hemoglobin
↓
impaired oxygen binding capacity

DIETARY IRON

- Primary form of iron: Fe^{+3} (ferric state) – not absorbable
- Stomach acid reduces Fe^{+3} to Fe^{+2} (ferrous state) – absorbable (however, only 10% of dietary intake is absorbed)
- Iron from plant sources (non-heme) is 3x less absorbable than animal sources (heme)
- Recommended daily dietary intake depends on a patient's age, sex, and medical history
    - General rule of thumb (average):
        - All males (19-50 years of age): 8 mg
        - Menstruating females (19-50 years of age): 18 mg
        - Pregnant females: 27 mg

IRON THERAPY

Oral supplementation is usually first-line

- Treatment may continue for 3-6 months once Hb normalizes, sometimes lifelong
- Adverse effects: primarily GI (take with food if intolerable effects, however, absorption is greater on empty stomach GI), constipation
- Liquid formulations cause tooth staining
- Polysaccharide iron complex (e.g. FeraMAX®) may be better tolerated, but less efficacious than ferrous salts
- Heme iron polypeptide (e.g. Proferrin®) may be better tolerated than iron salts, but lacks evidence for use in anemia. Bovine source

Parenteral iron therapy

- Due to high costs and inconvenient administration, parenteral iron is generally reserved for patients who are unresponsive to oral therapy, have chronic kidney disease, have had a gastric resection or chronic blood loss, have malabsorption syndromes (e.g. celiac disease), need a rapid increase in iron levels, etc.
- Can cause anaphylaxis and hypotension

ORAL IRON THERAPY

Oral iron	Dose (mg)	Elemental (mg)
Ferrous gluconate	300	35
Ferrous fumarate	300	100
Ferrous sulfate	300	60
Ferrous bisglycinate	25	25
Liposomal iron (Ferosom Forte®)	30 (in capsule form)	30
Heme iron polypeptide	11	11
Polysaccharide iron complex	150	150

INTERACTIONS AND ABSORPTION

- Vitamin C enhances absorption (not strong quality evidence)
- Calcium, antacids, tea, coffee, red wine decrease absorption
- Take iron salts on an empty stomach for optimal absorption (e.g. 1 hour before or 2 hours after a meal)
- Medications and other supplements that interact with iron include: cholestyramine, levodopa, sodium bicarbonate, fluroquinolones, tetracyclines, proton pump inhibitors
- Space out administration of iron salts with the above medications (time varies based on agent)
- Iron absorption may be improved if oral iron supplement given every other day or 1 dose daily (high amount of iron increases hepcidin levels and decreases iron absorption)

MACROCYTIC ANEMIA: MEGALOBLASTIC ANEMIA

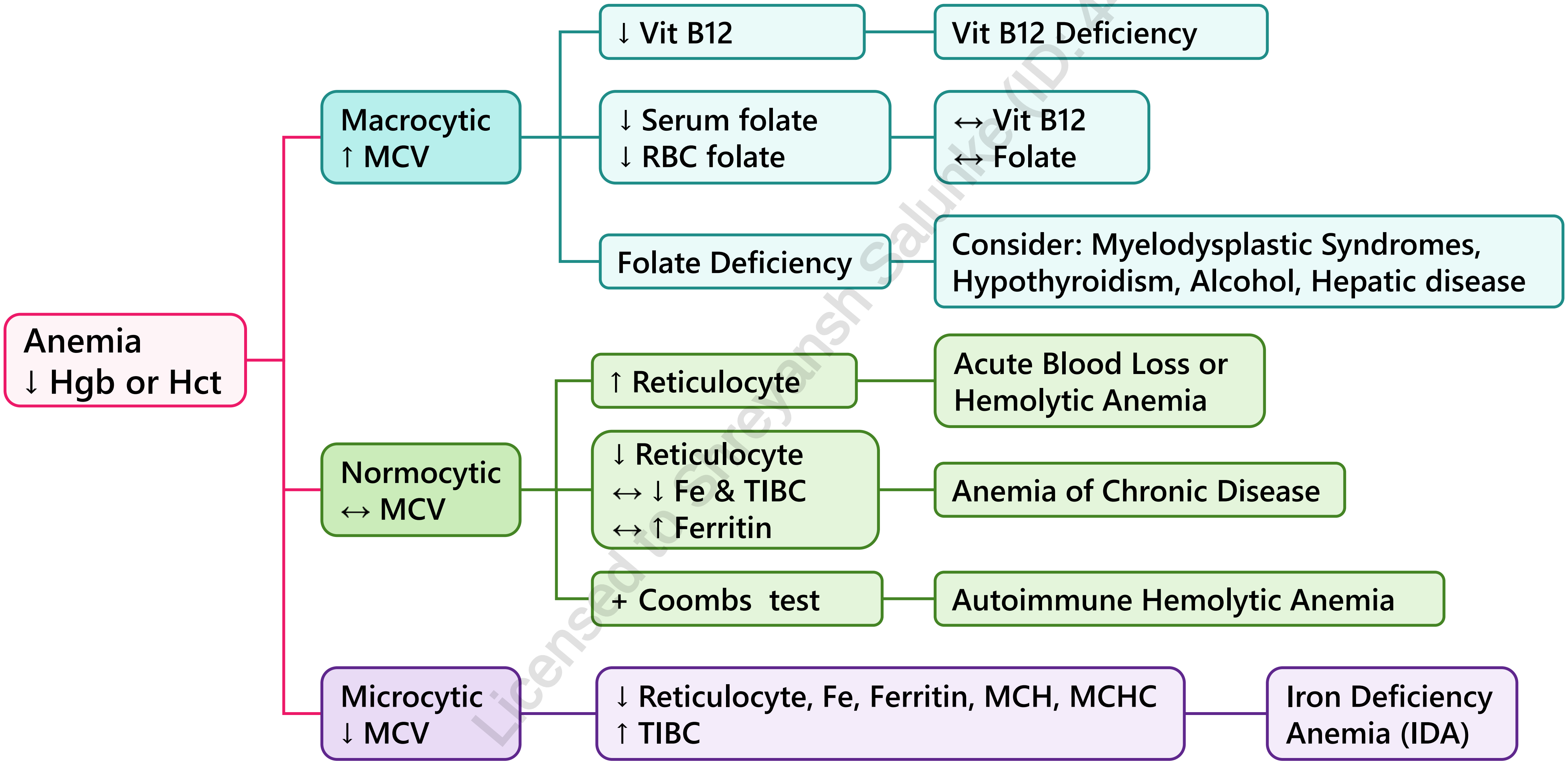
Causes	• Folate or vitamin B12 deficiency resulting in abnormally large RBCs and not enough normal RBCs
Lab values	• Increased MCV • Decreased RBC folate: folate-deficiency anemia • Decreased vitamin B12: vitamin B12-deficiency anemia

Folic acid and vitamin B12 are required for DNA and RNA synthesis

MACROCYTIC ANEMIA: MEGALOBLASTIC ANEMIA

	Folate-deficiency anemia	Vitamin B12-deficiency anemia
Common causes	• Inadequate intake: individuals with alcohol use disorder, chronically ill, elderly • Poor absorption: malabsorption syndromes • Increased requirements: pregnancy, hemolytic anemia, malignancy, chronic inflammatory disorders (e.g., rheumatoid arthritis) • Medications: folate antagonists (e.g., methotrexate), phenytoin (reduces absorption)	• Inadequate intake: individuals with alcohol use disorder, vegans, elderly • Decreased utilization: gastric bypass surgery, Crohn's disease • Intrinsic factor deficiency: pernicious anemia • Medications: metformin
Specific symptoms	Decreased appetite (weight loss), swollen red tongue, angular stomatitis, ecchymosis (bruising), neural tube defects (pregnancy)	Fatigue, headaches, palpitations, dyspnea, positive Babinski reflex, ataxia, vision loss, impaired proprio caption (positive Romberg's sign), impaired vibration perception
Drug therapy	• 1-5 mg/day of folic acid daily • Duration is typically up to 4 months (sometimes life-long) • Vitamin B12 deficiency must be ruled out before starting folic acid: folic acid supplementation can mask B12 deficiencies which allows the underlying neuropathy to persist	• Vitamin B12 deficiency must be treated as soon as possible due to possible neurological damage • Early treatment with cyanocobalamin is essential in preventing neurological damage • Oral dose: 1000-2000 mcg/day (monthly IM formulations also available) • Patients with pernicious anemia (intestinal B12 malabsorption) may require life-long therapy

ANEMIA ALGORITHM



MONITORING PARAMETERS

EFFICACY ENDPOINTS			
Parameter	**Degree of Change**	**Timeframe**	**Frequency of monitoring**
Hemoglobin	↑ by 10 g/L	7-10 days	Within 2-4 weeks of drug therapy initiation, then every 3 months for 1 year, then routinely thereafter
	Normalize to > 130 g/L in men and > 120 g/L in nonpregnant women	6-8 weeks	
Reticulocytes	23-90 x 10^9/L	Within 1 week	
B_{12} level	133-675 pmol/L	Start to rise within hours to days	Every 3-6 months
Serum folic acid levels	≥ 10 nmol/L	Start to rise > 2 weeks	
Neurologic deficits	Eradicate or reduce	≥ 6 months	At every appointment

MONITORING PARAMETERS

SAFETY ENDPOINTS		
Parameter	**Degree of Change**	**Timeframe**
Adverse effects of oral iron therapy (nausea, dyspepsia, constipation)	Prevent or minimize	Duration of therapy
Adverse effects of parenteral iron therapy (hypotension, fever, chills, headache, myalgia)		
Serious adverse effects of parenteral iron therapy (anaphylaxis)	Prevent	Duration of each infusion

TAKEAWAY

Macrocytic anemia: low B12, low folate or both
Microcytic anemia: low iron
Normocytic anemia: low RBC

- There is no evidence that one type of oral iron (e.g. ferrous sulphate, ferrous gluconate or ferrous fumarate) is more effective than another
- Heme iron polypeptide and polysaccharide iron complex: may be better tolerated but inconsistent reports in terms of efficacy

Duration of therapy:
- Oral iron should be taken for 3-6 months after target hemoglobin has been reached
- B12 and folate deficiencies: supplement until the deficiency is corrected

CASE 1

You are a pharmacist working in a family health team. You are consulted on the following case. HG is a 35-year-old female who presents to with fatigue and shortness of breath that has progressed in the last 2 months. She reports no symptoms of weakness or paresthesia. She also has heavy menstrual flow. She was previously seen at the clinic 6 months ago for fatigue and was started on ferrous gluconate 300 mg daily for presumed iron deficiency anemia (IDA). She takes the medication first thing in the morning. HG has a 3-year history of ulcerative colitis (UC), menorrhagia and gastroesophageal reflux disease (GERD). She takes Salofalk® (mesalamine) 1g TID, ibuprofen 200 mg po daily prn pain and esomeprazole 20mg once daily 30 minutes before breakfast. She is also on calcium 500 mg daily for general health. She last experienced a flare-up of UC 2 months ago and of GERD 1 year ago. She drinks 2 alcoholic drinks per day, 2 cups of coffee in the morning and 1 cup of black tea in the evening. She works full-time as a high school principal and occasionally uses cannabis as she finds that it helps relieve some of her stress. She follows a strict vegetarian diet. She has no known allergies to medications.
HG likes to go for walks in the morning and joins a yoga class on Sundays.

Her lab values are as follows:

- Hgb=111 (130-170 g/L)
- Hct= 0.35 (0.37-0.46 L/L)
- MCV=77.4 (80-100 fL)
- Ferritin= 205 (11-307 µg/L)
- Serum iron=3 (11-32 µmol/L)
- TIBC=90 (45-82 µmol/L)
- Transferrin saturation= 12% (25-40%)
- CRP=15 (<8 mg/L)
- Vitamin B12= 247 (133-674 pmol/L)
- Folate= 17 (>15 nmol/L)

CASE 1

1. Which of the following is LEAST likely to have contributed to the development of HG's anemia?
   a) Heavy menstruation
   b) Consumption of coffee and tea
   c) Ulcerative colitis
   d) Vegetarian diet

2. Which of these options represents the most appropriate first step in treating HG's anemia?
   a) Switching to ferrous fumarate 300mg po daily
   b) Consult her gastroenterologist for further treatment of her ulcerative colitis
   c) Switching to FeraMax® (polysaccharide iron complex) 150mg daily
   d) Counsel HG on optimizing the absorption of her ferrous gluconate

CASE 1

3. What is the most relevant advice you can provide to HG?
   a) Taper off esomeprazole if possible
   b) Reintroduce meat into her diet
   c) Take her ferrous gluconate with a glass of orange juice
   d) Switch to decaffeinated coffee

4. How long would you recommend HG to remain on iron supplementation after her hemoglobin normalizes?
   a) 3 months
   b) 6 months
   c) 12 months
   d) Indefinitely

REFERENCES

1. Schrier SL, Mentzer WC, Landaw SA. Approach to the adult patient with anemia. UpToDate.
2. WHO. Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity. Vitamin and Mineral Nutrition Information System. Geneva, World Health Organization, 2011 WHO/NMH/NHD/MNM/11.1 http://www.who.int/vmnis/indicators/haemoglobin.pdf.
3. DiPiro JT, Talbert RL, Yee GC, et al. Chapter 80.Anemias. *Pharmacotherapy: A Pathophysiologic Approach*. 9ed. McGraw-Hill: 2014
4. Lim W. Blood Disorders: Common Anemias. E-Therapeutics- Therapeutic Choices.
5. Dietary Reference Intakes tables: Reference values for elements. Updated December 18, 2023. Accessed March 4, 2025. Government of Canadahttps://www.canada.ca/en/health-canada/services/food-nutrition/healthy-eating/dietary-reference-intakes/tables/reference-values-elements.html.
6. Iron Storage Disease Training & Education. CDC Centers for Disease Control and Prevention. Atlanta GA. http://web.archive.org/web/20170325224948/http://www.cdc.gov/ncbddd/hemochromatosis/training/pathophysiology/iron_overload.html. Infufer Iron Dextran Injection USP Product Monograph. Sandoz Canada; Boucherville QC. file:///C:/Users/Owner/Downloads/PM00018721.PDF
7. Carr JH. 11689 Sickle Cell Anemia. PHIL CDC; 2009. Benz EJ. Clinical manifestations and diagnosis of the thalassemias. In *UpToDate*.
8. Mentzer WC. Hereditary spherocytosis: Clinical features, diagnosis, treatment. In *UpToDate*.
9. Vichinsky EP. Overview of the clinical manifestations of sickle cell disease. In *UpToDate*.
10. Schrier SL, Mentzer WC, Landaw SA. Approach to the adult patient with anemia. UpToDate. WHO. Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity. Vitamin and Mineral Nutrition Information System. Geneva, World Health Organization, 2011 WHO/NMH/NHD/MNM/11.1 http://www.who.int/vmnis/indicators/haemoglobin.pdf.
11. Zeller MP, Verhovsek M. Treating iron deficiency. *CMAJ* 2017 March 13;189:E409. doi: 10.1503/cmaj.160153
12. Practice Guidelines Iron Deficiency Anemia: Guidelines from the American Gastroenterological Association. American Academy of Family Physicians. 2021. file:///C:/Users/Fatimah%20Jaffer/Downloads/p211.pdf
13. Geneva: World Health Organization; 2024. Guideline on haemoglobin cutoffs to define anaemia in individuals and populations. https://www.who.int/publications/i/item/9789240088542
14. Iolascon A, Andolfo I, et. al. ; from EHA-SWG Red Cell and Iron. Recommendations for diagnosis, treatment, and prevention of iron deficiency and iron deficiency anemia. Hemasphere. 2024 Jul 15;8(7):e108. doi: 10.1002/hem3.108. PMID: 39011129; PMCID: PMC11247274.
15. Silverstein W, Cheung M, Lin Y. Vitamin B12 deficiency. Canadian Medical Association Journal. 2022. https://www.cmaj.ca/content/194/24/e843
16. Baddam S, Khan K, Jialal I. Folic Acid Deficiency. https://www.ncbi.nlm.nih.gov/books/NBK535377/

CHANGE LOG

July 2025
- Content reviewed and formatting changes made
- Slide 2: Updated Legend
- Slides 25 and 26: Updated case study and practice questions
- Slide 27: Added references

Aug 2025
- Formatting changes made throughout
- Added NAPRA competency slide 2

September/October 2025
- Minor formatting changes made
- Slides 25-26: Added monitoring parameter slides

January 2026
- Slide 28: Added more information to the case
- Slide 30: Removed 'non-pharmacological' from question

March 2026
- Slide 20: Added liposomal iron and ferrous bisglycinate